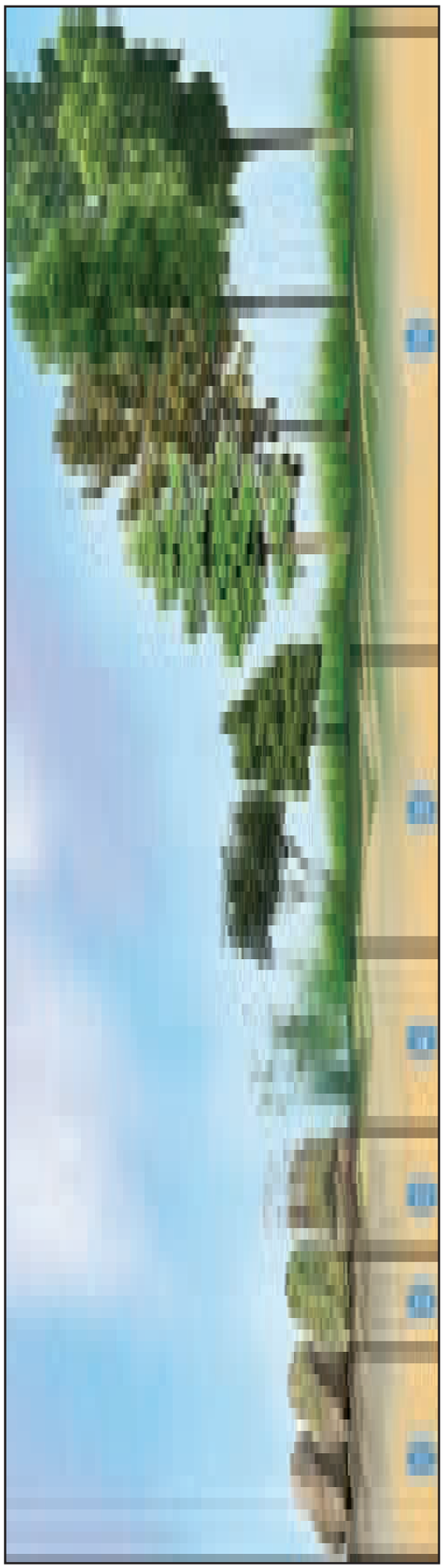


Figure 7.22: Diagrammatic representation shows different stages of lithosere

weathering of rock. Due to this continuous process, small particles of rocks are formed, which together with decaying lichen make the first thin layer of soil on rock surface. However, this process is very slow. At the end, this habitats become less fit for existing plants and is gradually replaced by another type of lichens called foliose lichen.

2. Foliose lichen stage - Gradually crustose lichens are replaced by foliose lichen like *Parmelia* and *Dermatocarpon* etc. These plants have leaf like structures. They also secrete some acids which further loosen the rocks into small soil particles. This process enhances water retaining capacity of the habitat and causes further accumulation of soil particles and humus. Gradual changes make the area less favourable for existing foliose lichen.

3. Moss stage - When the habitat is changed, the existing foliose lichen starts disappearing and favours the growth of some xerophytic moss like *Polytrichum, Tortula* and *Grimmia*. The luxurious growth of moss competes with lichens. Due to the death and decay of mosses, further addition of humus and moisture to the habitat takes place. Therefore, the next seral community tries to replace the moss community.

4. Herb stage - With the gradual disappearance of moss stage, herbaceous plant communities like *Aristida, Festuca* and *Poa,* etc., invade the habitat. The extensive growth of these herbs alter the habitat. The decaying leaves, stems, root and other parts of these plants get deposited on the soil surface in the form of humus. It further increases the water holding capacity of soil. These conditions become more suitable for shrubs.

5. Shrub stage - The habitat change results in the invasion of shrubs like *Rhus, Zizyphus, Capparis* and dominated by herbaceous plants. The death and decaying of

TN_GOVT_BOTANY_XII_PAGES_162-190 CH 07.indd 167 3/8/2020 2:54:06 PM